

# Shamit Sai

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## EDUCATION

**BITS PILANI, GOA CAMPUS**  
**BE IN ELECTRONICS AND**  
**COMMUNICATIONS ENGINEERING**  
**MINOR IN DATA SCIENCE**  
May 2027 | Goa  
CGPA: 8.3/10

## LINKS

Github:// **Light1224**  
LinkedIn:// **Shamit Sai**  
Codeforces:// **lightcf12**  
Blog and Portfolio:// **shamit**

## RELEVANT

## COURSEWORK

### MINOR

Machine Learning  
Foundations of Data Science  
Applied Statistical Methods  
Nonlinear Optimization

### UNDERGRADUATE

Computer Architecture  
Communication Networks  
Object Oriented Programming  
Information Theory and Coding  
Signals and Systems  
Microprocessors and Interfacing  
Probability and Statistics

## SKILLS

### PROGRAMMING

Proficient  
C++ • Shell • Python  
Intermediate  
Java • Assembly x86 • Git  
Coursework  
Vivado Xilinx • MATLAB

## COMPETITIVE

## EXAMINATIONS

JEE Mains - 99.1 percentile  
JEE Advanced - rank 5811  
KCET - rank 759  
ComedK - Rank 57  
12th Boards - 95/100

## PROJECTS

### DC-UTPX Jan 2026 – Feb 2026 | Independent

- Designed and implemented a userspace UDP transport protocol for data-center incast workloads in C++, combining DCTCP-style ECN feedback, delay/loss congestion control, and microburst pacing; achieved 99.97% delivery ratio and P99 RTT < 2.2 ms across 73 concurrent flows under 16-sender fan-in stress.
- Outperformed kernel TCP by 9× in aggregate throughput (856 Mbps vs. 95 Mbps) under synchronized incast and by 6.6× under queue-shaped buffer-sweep scenarios, validating that adaptive userspace congestion/pacing substantially reduces head-of-line blocking in multi-sender data-center traffic patterns.
- Instrumented the protocol stack with eBPF-based per-flow telemetry, capturing RTT percentiles (P50–P99.9), congestion window trajectories, ECN-mark fractions, and drop-ratio signals at sub-millisecond granularity across incast, mixed elephant-mice, and buffer-sweep workload families.
- maintained Jain fairness index  $\geq 0.945$  and goodput loss < 0.003% of sent traffic across all three workload scenarios

### GEMM KERNEL Dec 2025 – Jan 2026 |Independent

- Built a C++20 GEMM kernel on Apple M2 progressing from naive to cache-blocked, NEON-SIMD, packed, and multithreaded stages, achieving **231 GFLOP/s** (95% of OpenBLAS) — a **223× speedup** over baseline.
- Implemented optimized 8×8 ARM NEON FMA microkernels, increasing performance from 16 to **42.7 GFLOP/s** by fully utilizing SIMD units.
- Engineered an L2-tuned, SIMD-aligned memory packing system eliminating dynamic allocation in hot paths, delivering a **2.84× throughput gain**.
- Parallelized across 8 cores using `std::thread`, scaling from 43 to **231 GFLOP/s** with near-linear speedup.

## EXPERIENCE

### SRLDC - GOVT OF KARNATAKA | SUMMER INTERN

May 2025 – July 2025 | Bengaluru, Karnataka

- Designed and benchmarked production-grade time-series forecasting models (NNs, gradient-boosted trees, ensembles) for next-day grid demand prediction, achieving **92%** accuracy and outperforming vendor forecasts.
- Engineered advanced temporal and weather-derived features (lag structures, rolling statistics) to improve robustness and predictive performance
- Built and deployed a Linux-based inference pipeline consuming minute-level PostgreSQL data, automated via `cron` to generate rolling forecasts every 15 minutes (24/7 operation).
- Integrated forecasts and evaluation metrics into Grafana dashboards for real-time monitoring and operational visibility.

## COMPETITIVE PROGRAMMING

|         |                      |            |
|---------|----------------------|------------|
| Current | peak rated 1547      | Codeforces |
| 2025    | Solved 100+ problems | LeetCode   |